

1. (3 points) A statistics department at a large university gives students the option of three majors. 30% of the department's students choose to major in mathematical statistics, 15% in biostatistics, and the remaining 55% in data science. Suppose that 30% of those in mathematical statistics go on to graduate school, as opposed to 20% for those in biostatistics and 10% for those in data science. Find the probability that a randomly chosen alum who attended graduate school majored in data science. Show your work.

Solution: $P(DS|GS) = \frac{.55*.1}{.55*.1 + .15*.2 + .3*.3} = .3143$

2. (9 points) Consider a parameter of interest, θ , that can only take on the values 0, 0.05, and 0.1. The table below shows a prior distribution for θ .

θ	0	0.05	0.1
$f(\theta)$	.7	.2	.1

We observe a vector of independent observations $Y_1, \dots, Y_n$ with pmf given by the table below.

y	1	3	7
$f(y \theta)$	$0.4 + \theta$	$0.3 + \theta$	$0.3 - 2\theta$

- (a) Suppose we have three observations: $Y_1 = 3, Y_2 = 3, Y_3 = 7$. Calculate the likelihood $L(\theta = .05|Y_1 = 3, Y_2 = 3, Y_3 = 7)$. Show your work.

Solution: $L(\theta = .05|y) = (.35)(.35)(.2) = .0245$

- (b) Show that $f(Y_1 = 3, Y_2 = 3, Y_3 = 7) = .027$. (Hint: Calculate the product of the prior and likelihood for each θ then take the sum.)

Solution: $f(Y_1 = 3, Y_2 = 3, Y_3 = 7) = f(\theta = 0)f(Y_1 = 3, Y_2 = 3, Y_3 = 7|\theta = 0) + f(\theta = .05)f(Y_1 = 3, Y_2 = 3, Y_3 = 7|\theta = .05) + f(\theta = .10)f(Y_1 = 3, Y_2 = 3, Y_3 = 7|\theta = .10) = .7 * .3 * .3 * .3 + .2 * .35 * .35 * .2 + .1 * .4 * .4 * .2 = .0302$

- (c) Using (a) and (b), calculate the posterior probability $f(\theta = 0.05|Y_1 = 3, Y_2 = 3, Y_3 = 7)$.

Solution: $f(\theta)L(\theta = .05|y) = .2 * .0245 / .027 = .1815$

3. (10 points) A softball team wants to build a Gamma-Poisson model for λ , the average number of runs it scores per game. It asks two student data analysts to give prior distributions of λ . Leticia suggests a Gamma(12,2) prior, while Emmalee suggests a Gamma(24,4) prior.

- (a) The team has a new coach this year that plans to play a different style. She advises her student analysts to go with the more vague prior. In the context of this problem, what is meant by a “vague prior”?

Solution: A vague prior in this case is one that is not very confident in the true average number of runs that will be scored per game.

- (b) Calculate the standard deviation for both priors. Use those values to determine the prior that is more vague. Explain your reasoning.

Solution: The Gamma(12,2) has a standard deviation of $\sqrt{12/4} = 1.73$. The Gamma(24,4) has a standard deviation of $\sqrt{24/16} = 1.22$. Thus the Gamma(12,2) prior is more vague because it has a larger standard deviation.

- (c) In the first five games of the season they scored a total of 22 runs. Using this data and the prior from part (b), give the posterior distribution of λ .

Solution: Gamma(34,7)

4. (3 points) Consider a parameter π with prior distribution $\pi \sim \text{Beta}(\alpha, \beta)$. The conditional distribution of the data Y is $Y|\pi \sim \text{Binomial}(n, \pi)$.

- (a) Write the formula for the posterior mode as a function of α, β, y , and n .

Solution: $\frac{\alpha+y-1}{\alpha+\beta+n-2}$

- (b) (Bonus) Show that the posterior mode can be written as a weighted average of the prior mode and observed sample proportion.

Solution:
$$\begin{aligned} \frac{\alpha+y-1}{\alpha+\beta+n-2} &= \frac{\alpha-1}{\alpha+\beta+n-2} + \frac{y}{\alpha+\beta+n-2} = \frac{\alpha-1}{\alpha+\beta+n-2} \frac{\alpha+\beta-2}{\alpha+\beta-2} + \frac{y}{\alpha+\beta+n-2} \frac{n}{n} \\ &= \frac{\alpha+\beta-2}{\alpha+\beta+n-2} \frac{\alpha-1}{\alpha+\beta-2} + \frac{n}{\alpha+\beta+n-2} \frac{y}{n} \end{aligned}$$